Functions & Graphs

Evaluating functions

1 If $f(x) = 3x^2 - 2x + 5$, find $f(-2)$.

2 If $g(x) = \frac{2x+1}{x-3}$, find $g(5)$.

Domain and range

3 Find the domain of $f(x) = \frac{4}{x+7}$.

4 Find the domain of $g(x) = \sqrt{2x-8}$.

Composite functions

5 If $f(x) = 2x + 1$ and $g(x) = x^2 - 3$, find $f(g(2))$.

6 If $f(x) = x + 4$ and $g(x) = 3x$, find $(f \circ g)(x)$ and $(g \circ f)(x)$, and explain why they differ.

Graphing linear and quadratic functions

7 Find the x- and y-intercepts of $y = 2x - 6$.

8 Find the vertex of $y = x^2 - 8x + 15$.

Transformations of graphs

9 The graph of $y = x^2$ is shifted 4 units left and 3 units down. Write the equation of the resulting graph.

10 Describe how the graph of $y = -3(x-2)^2 + 1$ differs from the graph of $y = x^2$.

Piecewise and step functions

11 A function is defined as $f(x) = 2x$ for $x \leq 2$ and $f(x) = x + 5$ for $x > 2$. Find $f(0)$ and $f(4)$.

Interpreting graphs in context

12 A graph shows a car's speed over time: the speed increases from 0 to 40 mph over the first 10 seconds, stays constant at 40 mph for the next 20 seconds, then decreases to 0 over the final 10 seconds. Describe what is happening to the car during each phase.

Finding the equation of a function from a graph description

13 A parabola has x-intercepts at $x = -1$ and $x = 5$, and passes through $(2, -9)$. Find its equation in factored form.

Multi-part function analysis

14 This question has two parts. A function is given by $f(x) = x^2 - 4x - 5$. (a) Find the zeros of the function by factoring. (b) Find the vertex, and state whether it is a minimum or maximum.

Inverse functions

15 Find the inverse of $f(x) = 4x + 8$.

16 If $f(3) = 10$, find $f^{-1}(10)$, using the definition of an inverse function.

Rational and radical functions

17 Find the vertical asymptote of $f(x) = \frac{3}{x - 5}$.

18 Find $g(9)$ for $g(x) = \sqrt{x} + 4$.

Exponential functions

19 If $f(x) = 2^x$, find $f(5)$ and $f(-2)$.

20 A function is given by $g(x) = 100(0.5)^x$. Find $g(3)$ and interpret its meaning if this models a decaying quantity over x time periods.

Multi-part analysis of a real-world function model

21 This question has two parts. A ball is thrown, and its height is modeled by $h(t) = -16t^2 + 48t + 4$, where t is time in seconds. (a) Find the height of the ball at $t = 0$ and interpret this value. (b) Find the time at which the ball reaches its maximum height, and find that maximum height.